

Stability of equilibria and Linear ODEs

1. *What does stability mean?* Let $\bar{x}$ be an equilibrium of autonomous ODE $\dot{x} = F(x)$ where F is C^1 everywhere.

Definition 9.1 (Stability). The equilibrium $\bar{x}$ is **stable** if

$$\forall \varepsilon > 0 \quad \underline{\hspace{2cm}}$$

such that every solution starting in $B_\delta(\bar{x})$ satisfies

$$\underline{\hspace{10cm}}.$$

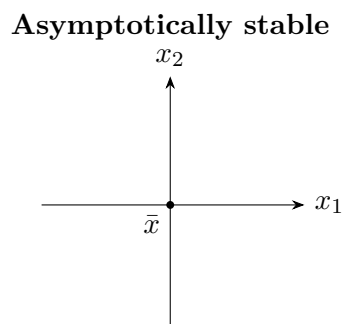
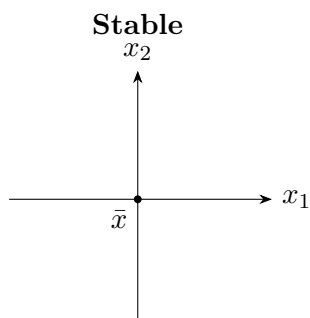
It is **unstable** if it is $\underline{\hspace{2cm}}$.

Definition 9.2 (Asymptotic stability). The equilibrium $\bar{x}$ is **asymptotically stable** if it is $\underline{\hspace{2cm}}$ and there exists $r > 0$ such that

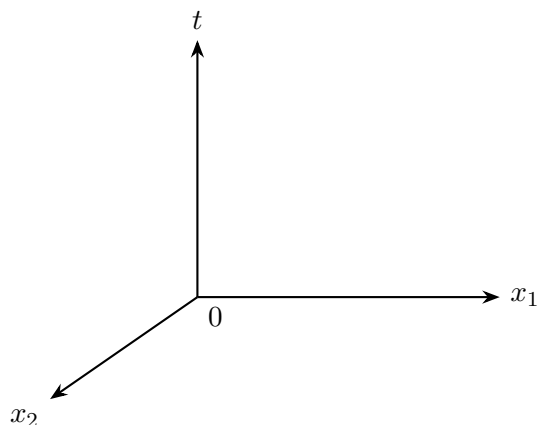
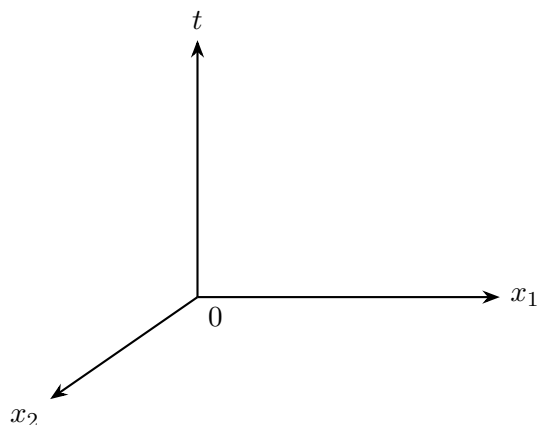
$$x(0) \in B_r(\bar{x}) \implies \underline{\hspace{10cm}}.$$

Sketch and label the two ideas.

On the left, show an ε -neighbourhood, a smaller δ -neighbourhood, and an orbit that stays within the ε -neighbourhood. On the right, show a nearby orbit that also converges.



Sketch and label the state-trajectories for these two ideas. Does stability alone imply convergence?



2. Scalar equations: read the direction of motion

Consider $\dot{y} = f(y)$ with $f \in C^1$. Equilibria are the zeros of _____.

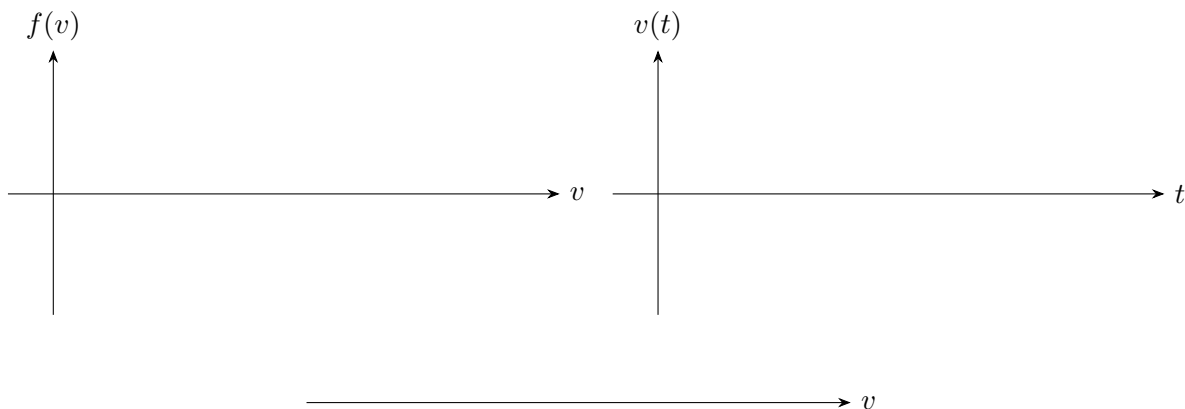
$$f(y) > 0 \Rightarrow y(t) \text{ is } \underline{\hspace{2cm}}, \quad f(y) < 0 \Rightarrow y(t) \text{ is } \underline{\hspace{2cm}}.$$

Example 9.1 – RC circuit. Let $R, C > 0$ and let the source voltage V_0 be constant:

$$\dot{v} = -\frac{1}{RC}v + \frac{V_0}{RC} = f(v).$$

(a) Find the equilibrium and determine the sign of f on either side.

(b) Sketch $f(v)$ for $V_0 > 0$, then transfer the signs to a vector field and trajectory.



(c) Classify the equilibrium.

What should a scalar stability test check near an equilibrium?

3. A nonlinear scalar ODE and its trajectories

Example 9.2 – Three equilibria. Consider $\dot{y} = y(y - 1)(4 - y)$.

(a) Find all equilibria. Complete the signs and add direction arrows.

Equilibria: _____.

Interval	$(-\infty, 0)$	$(0, 1)$	$(1, 4)$	$(4, \infty)$
Sign of $f(y)$	_____	_____	_____	_____



(b) Classify each equilibrium. Justify the classification using the arrows.

(c) Sketch $y(t)$ for initial values in each of the four intervals.

Include the equilibrium solutions and indicate the limits as $t \rightarrow \infty$.



(d) Why can a trajectory not cross an equilibrium?

Example 9.3 – A one-sided attraction is not enough. For $\dot{y} = y^2$, draw the phase line near 0 and classify the equilibrium.

4. A precise scalar stability criterion

Consider $\dot{y} = f(y)$ with $f \in C^1$ on an open interval containing $\bar{y}$. An equilibrium is **isolated** if some neighbourhood of it contains _____.

Theorem 9.3 (Scalar asymptotic stability). Suppose $\bar{y}$ is an isolated equilibrium. Then $\bar{y}$ is asymptotically stable if and only if there exists $r > 0$ such that

$$\text{_____} \quad \text{whenever } 0 < |y - \bar{y}| < r.$$

Equivalently, near $\bar{y}$,

$$y < \bar{y} \Rightarrow f(y) \text{ _____}, \quad y > \bar{y} \Rightarrow f(y) \text{ _____}.$$

Remark (The role of isolation). For an isolated equilibrium of a scalar C^1 autonomous ODE,

$$\text{stable} \implies \text{_____}.$$

Why is the word “isolated” essential? Consider $\dot{y} = 0$.